浙江大学 2023–2024 学年秋冬学期

《计算机组成与设计》课程期末考试试卷

课程号：111C0130，开课学院：信息与电子工程学院

考试试卷：A 卷、√B 卷（请在选定项上打√）

考试形式：√闭、开卷（请在选定项上打√），

允许带 1 张 A4 纸大小的手写资料和计算器入场

考试日期：2024 年 1 月 16 日，考试时间：120 分钟

诚信考试，沉着应考，杜绝违纪。

考生姓名：_____学号：_____所属院系（专业）：_____

题序	一	二	三	四	五	总 分
得分						
评卷人						

一．**CHOICE** (60 points): (note: only **one** is correct)

2^26

1. Assume that a virtual memory address is 32bit in length and the physical memory is 64MB. What's the size of a page table if page size is 32KB? (D)

2^19

A: 32KB B: 64KB 2^15 C: 128KB 1KB D: 512KB

offset=15bits Pentry=2^11 Pblock=16=2^4个

2. Dividing 10101_2 by 11_2, the quotient is ___and the reminder is___ . C

A. 100_2 01_2

B. 111_2 01_2

C. 111_2 0_2

D. 100_2 0_2

8bytes

3. A direct-mapped cache has 1K entries, cache line size is 2 words. What's the capacity of the cache? (C)

A: 2KB B: 4KB C: 8KB D: 16KB

4. For a direct-mapped cache design with 32-bit address, the following bits of the address are used to access the cache.

Tag	Index	Offset
31-12	11-15	4-0

a) What is the cache line size (in words)? (A)

block size=2^offset=2^5=32

A：8 B：4 C：16 D：2

b) How many entries does the cache have? (B)

entry=2^index=2^5

A：128 B：32 C：64 D：16

5. Assume that integer is stored as a word (4 bytes) in memory, and -2 is stored at Memory address start from 0x00000004, so what Byte is store in memory address 0x00000007 in RISC-V

computer? (A)
A. 11111110 B. 11111111 C. 10000000 D. 00000010

6. Cache acts between _____. A
A. CPU and RAM
B. RAM and ROM
C. CPU and Hard disk
D. none of these

7. MIMD stands for ____. A
A. Multiple instruction multiple data
B. Multiple instruction memory data
C. Memory instruction multiple data
D. Multiple information memory data

8. Which of the following cache designer guideline is **not valid**? (B)
A. Fully associative caches have no conflict misses.
B. In reducing misses, associativity is more important than capacity.
C. The higher the memory bandwidth, the larger the cache block.
D. The shorter the memory latency, the smaller the cache block.

9. The idea of cache memory is based on _____. A
A. the property of locality of reference
B. the heuristic 90-10 rule
C. the fact that reference generally tend to cluster
D. all of the above

10. Which one is not one of the five classic components of a computer? (B)
A：Input B：Bus C：Memory D：Output

11. Suppose there was a 16-bit IEEE 754 floating-point format with five exponent bits. What would
16-5-1=10bits of fraction
be the likely range of numbers it could represent? (C)
A. 1.0000 0000 00×2_0 to 1.1111 1111 11×2_{31},0
B. ±1.0000 0000 0×2_{-14} to ±1.1111 1111 1×2_{15}, ±0, ±∞, NaN
C. ±1.0000 0000 00×2_{-14} to ±1.1111 1111 11×2_{15}, ±0, ±∞, NaN
D. ±1.0000 0000 00×2_{-15} to ±1.1111 1111 11×2_{14}, ±0, ±∞, NaN

12. ____ register keeps track of the instructions stored in program stored in memory. C
A. AR (Address Register)
B. XR (Index Register)
C. PC (Program Counter)
D. AC (Accumulator)

13. Assume the current PC is 0x10000010, what's next value of PC after execution of "jal x0, 100"? (C)

A.0x00000064 B. 0x10000014 C. 0x10000074 D. 0x10000064

14. The fastest data access is provided using (D).

A. Caches B. DRAM's C. SRAM's D.Registers

15. Assuming that there are 100 processors, what is the proportion of parallel part in parallel software to achieve a 80 speedup? (A)

A：99.75% B：99.15% C：98.75% D：98.15%

16. Assuming that individual stages of the datapath have the following latencies.

IF	ID	EX	MEM	WB
200ps	150ps	120ps	190ps	140ps

a) What is the clock cycle time in a pipelined and nonpipelined processor? (D)

A：200ps, 200ps B：120ps, 200ps C：120ps, 800ps D：200ps, 800ps

b) What is the total latency of a ld instruction in a pipelined and nonpipelined processor? (C)

A：1000ps, 200ps B：800ps, 800ps C：1000ps, 800ps D：200ps, 200ps

17. Which one of three categories occurs in a set-associative or direct-mapped cache when multiple blocks compete for the same set and that are eliminated in a fully associative cache of the same size? (C)

A：compulsory misses B：capacity misses C：conflict misses

18. Assuming forgetting to implement the hazard detection unit, how many cycles does it take to execute the following instruction sequence on a 5-stage pipeline with CPI 1? (C)

```
I1:  ld   x1,40(x6)
I2:  add  x2,x3,x1
I3:  ld   x1,16(x2)
I4:  sd   x2,20(x4)
I5:  and  x1,x1,x4
```

A, 8 B：7 C：6 D：5

二. **TRUE (T) OR FALSE (F)** (10 points)

[T] (1) A virtual cache access time is always faster than that of a physical cache?

[F] (2) High associativity in a cache reduces compulsory misses.

[F] (3) A write-through cache typically requires less bus bandwidth than a write-back cache.

[F] (4) Cache performance is of less importance in faster processors because the processor speed compensates for the high memory access time.

[T] (5) Memory interleaving is a technique for reducing memory access time through increased bandwidth utilization of the data bus.

[F] (6) The direct-mapped cache has largest tag overhead.

[T] (7) Pipelining improves performance by increasing instruction throughput.

[T] (8) Cache and main memory are made of SRAM and DRAM respectively.

[T] (9) In a cache system, write allocate policy always coordinates with write back policy.

[F] (10) We use large cache line due to temporal locality.

三. **PIPELINE** (10 points)

A certain workload having the following instruction mix is run on two processor designs with both having I-Cache and D-Cache.

ADD	10%
AND	20%
BEQ	25%
SD	15%
LD	30%

1/200MHz=5ns
75ns/5ns=15cycles
1+0.1*5+0.02*0.45*15+0.55*0.25*3+0.25*0.3*1

每次未被采纳的分支需要三个周期

Additionally, it is known that I-Cache Hit-rate is 90%, D-Cache Hit-rate is 98%, 45% branches are not taken and 25% of LD instructions are followed by a dependent instruction. The memory takes **75 nano-seconds** to access.

依赖引入一个额外周期

1. Assuming the above code is run on a standard 5-stage pipeline design processor with forwarding and hazard detection and with branches resolved in MEM and predicted not taken and clocked at 200MHz, what is the CPI? [4 points]

Clock period : 1 / 200MHz = 5ns

Cache : 75 ns / 5ns = 15 cycles

CPI = 1 + 1*0.10*15 + (0.3+0.15)*0.02*15 + 0.3*0.25*1 + 0.25*0.55*3 = 3.1225

D-cache未命中（涉及ld、sd）

I-cache未命中

每次未命中需要15个额外周期

1/400MHz=2.5ns,75ns/2.5ns=30cycles
1+0.1*30+0.02*0.45*30+0.55*0.25*10+0.25*0.3*1

2. This five stage pipeline is extended to a similar 15 stage pipeline with no additional hazards being introduced. The amount of stall cycles needed for a ld followed by a dependent instruction does not change. The new frequency is 400MHz. Now the same code is run on the 15 stage pipeline where branches are resolved in the 11th stage.

2.1 What is the new CPI? Show your work. [4 points]

2.2 Does this new design result in better performance for this workload? [2 points]

a)：

Beq : 11 – 1 = 10 cycles

Clock period : 1/400 MHz = 2.5ns

Cache : 75ns / 2.5 ns = 30 cycles

CPI = 1 + 1*0.1*30 + (0.3+0.15)*0.02*30 + 0.3*0.25*1 + 0.25*0.55*10 = 5.72

I-cache miss　D-cache miss　ld依赖　branch未执行

b)：

first : 5 ns * 3.1225 = 15.6125 ns ;

second : 2.5 ns * 5.72 = 14.3 ns ;

Yes, this new design result in better performance for this workload.

四．CACHE (10 points)

For all the questions in this problem, assume that we are using a 12-bit machine with a byte-addressable memory and a direct-mapped cache. The cache can hold up to 16 cache lines.

entry=16

1.How many bits do you need for the set index? (2pts) 4

_____ index=4

2. The following sequence of 9 memory accesses generates the hits/misses shown. Some miss/hit entries are intentionally left blank. The cache is initially empty. Note that the addresses are written in binary with spaces added between each 4 bits for readability. These are not necessarily the tag/index/offset boundaries.

index=4

#	Address	Hit/Miss
1	1101 1111 0000	Miss
2	0000 1101 1111	Miss
3	1101 0111 0101	Miss
4	0000 1101 1100	Hit
5	1101 1111 0011	Miss
6	1111 0111 0010	Miss
7	1101 1111 0000	Miss
8	0000 1101 1101	Hit
9	1111 0111 0100	Miss

offset<2
offset>=2

2.1 What is the number of tag bits? (2pts) 5

_____

2.2 What is the number of offset bits? (2pts) 3

_____

2.3 Fill the miss/hit for each of the blank entries. (4pts)

五．VIRTUAL MEMORY (10 points)

Assume the following: a 32-bit virtual address space, with a 1KB page size

1. How many bits are in the offset portion of the virtual address? (2 points)

_____ 10 offset=10

2. How many bits are in the VPN portion of the virtual address? (2 points)

_____ 22 32-10=22

s

Now, let's focus on the page table. Assume each page table entry is 4 bytes in size. Assuming a linear page table:

3. How many entries are in the table? (2 points)

_____ 2^22 entry=2^VPNsize=2^22

4. What is the total size of the table? (2 points)

_____MB 16 PTsize=2^22*4byte=2^24byte=16MB

5. In a live system, how much memory would be occupied by page tables if there are 100 processes? (2 points)s

_____MB 1600